

STATATHON 2025



- Problem Statement ID: **PS-5**
- Problem Statement Title: **AI-enabled Semantic Search for National Classification of Occupation (NCO)**
- PS Category- Software/Hardware: **Software**
- Team ID: **5252**
- Team Name (Registered on portal): **StataStorm**

AI-enabled semantic search for National Classification of Occupation (NCO)

Detailed Explanation

- **Data Preparation:** Convert NCO-2015 dataset into structured format (JSON/CSV) while preserving hierarchy from division to 8-digit occupation codes.
- **Semantic Search Engine:** Use pre-trained NLP models (Sentence Transformers) to generate embeddings for job titles & descriptions.
- **Fast Retrieval:** Store embeddings in a vector database (FAISS) for instant, ranked search results.
- **User-Friendly Interface: Simple web or mobile UI with:**
 - Text & Voice input (English & Hindi)
 - Top N matches with confidence scores
 - Occupation hierarchy visualization
- **Smart Matching:** Built-in synonym & variation mapping (e.g., “tailor” ↔ “sewing machine operator”).

How It Addresses the Problem

- **No More Guesswork:** Removes need to memorize NCO structure.
- **Time Saver:** Reduces code search time from minutes to seconds.
- **Error Reduction:** Improves accuracy of classification, ensuring high-quality official statistics.
- **Accessible Anywhere:** Works offline for rural or low-connectivity areas.

Innovation

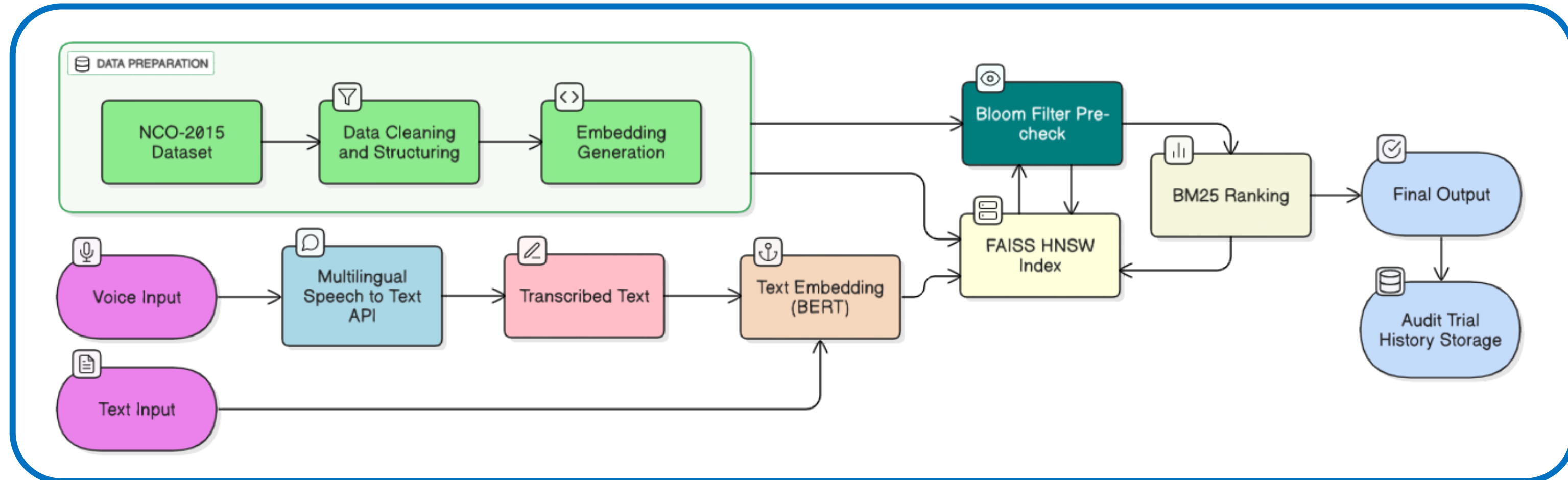
- **Multilingual + Voice Support** for field usability and inclusivity
- **Explainable Results** - Confidence scores & “Why this match” tooltips for transparency.
- **Accelerated Semantic Search** – Uses Bloom Filter–based pre-filtering with BM25 ranking, combined with FAISS’s HNSW index, to deliver ultra-fast and scalable occupation code retrieval.

TECHNICAL APPROACH

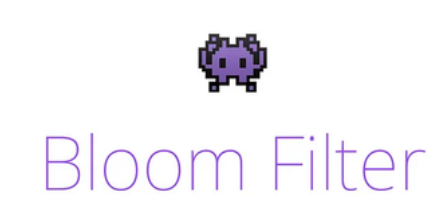
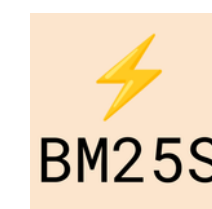


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Architecture



Technical Stack



Feasibility Analysis

Dataset Ready: NCO-2015 provided by MoSPI in usable formats.

Proven Tech Stack: Pre-trained NLP models + FAISS HNSW index ensure fast semantic search.

User-Friendly: Simple web/voice interface usable by survey staff without training.

Low Cost: Open-source tools, minimal hardware, offline-capable.

Scalable: Easily extendable to more languages and other classification systems.

Potential Challenges & Risks

Multilingual Accuracy: Hindi/regional language queries may produce weaker results.

Large Dataset Search Speed: Full NCO search may slow down without optimization.

Synonym Handling: Variations in job titles might cause missed matches.

Overcoming Challenges

Multilingual Accuracy: Preload common Hindi job titles and use a lightweight translation API for others.

Search Speed: Combine Bloom Filter pre-checks + BM25 ranking with FAISS HNSW index for sub-second search times.

Synonym Handling: Since we use dense embeddings from a BERT-based model, similar job titles (e.g., “tailor” and “sewing machine operator”) generate similar vector representations, enabling the semantic search engine to match them accurately without needing exact keywords.

Potential impact

Faster Classification: Reduces occupation code search time from minutes to seconds.

Higher Accuracy: Minimizes errors in classification, improving quality of official statistics.

Field Usability: Works for multilingual and voice inputs, helping rural and less digitally-trained staff.

Scalable Nationwide: Can be deployed across all MoSPI surveys and extended to other classification systems

Benefits

Policy Enablement: Timely, accurate data supports better government planning and targeted programs.

Environmental Sustainability: Reduces paper use by replacing printed manuals with digital tools.

Economic Impact: Lower manpower costs, reduced training time, and quicker survey completion.

Technological Advancement: Brings AI-powered search to official government workflows.

Scalability & Reusability: Can be adapted to other classification systems like ISCO or sector-specific taxonomies.

1. National Classification of Occupations (NCO-2015) - Ministry of Labour & Employment, Government of India
2. Sentence Transformers Documentation - Pre-trained models for semantic search
3. FAISS – Facebook AI Similarity Search - Vector database library
4. BM25 Ranking Algorithm - Okapi BM25 overview
5. Bloom Filter - Probabilistic data structure for fast membership tests
6. Semantic Similarity Strategies for Job Title Classification
Discusses using word embeddings (like Word2Vec and Word Mover's Distance) to classify job titles into predefined categories effectively
7. Bias in Occupation Classification via Semantic Representation
Highlights potential gender bias in semantic classification systems